

Mach-Zehnder Interferometer Performance Study

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Abstract—The transmission spectrum of a silicon-based Mach-Zehnder Interferometer (MZI) is numerically simulated across multiple parameter configurations using the Finite-difference eigenmode (FDE) solver in Lumerical MODE and INTERCONNECT circuit simulations. The study examines how key design parameters including arm length difference, waveguide width, beam splitter coupling ratio, and operating temperature influence the spectral response and output intensity characteristics of the device. Compact waveguide models are derived and validated against simulation data, and a method for extracting the waveguide group index directly from the free spectral range is demonstrated. The fabricated MZI3 device was characterized experimentally, yielding a measured group index of $n_g = 4.2179$ in agreement with corner analysis predictions.

Keywords—Mach-Zehnder Interferometer, strip waveguide, free spectral range, quasi-TE mode, Lumerical INTERCONNECT, thermo-optic effect, FDE mode solver, group index, optical splitting ratio.

I. INTRODUCTION

THE Mach-Zehnder Interferometer (MZI) is one of the most fundamental and widely adopted building blocks in silicon photonic integrated circuits (PICs). Owing to its compatibility with standard CMOS fabrication processes and its ability to precisely control light interference, the silicon-based MZI has attracted considerable research interest. Its versatility has enabled a broad range of applications, including high-speed optical modulators, reconfigurable optical switches, and wavelength-selective optical routers [1], [2].

In this report, the transmission spectrum of silicon-based MZI structures is numerically simulated, experimentally fabricated, and characterized across several design configurations. The investigation is structured in six stages. In the first stage, a standard silicon-on-insulator (SOI) waveguide with a rectangular cross-section of $500 \times 220 \text{ nm}^2$ is considered. The fundamental quasi-transverse-electric (quasi-TE) mode is selected for propagation analysis, and the MZI performance is evaluated under both balanced and imbalanced arm configurations. In the second stage, the waveguide cross-section is modified to $600 \times 220 \text{ nm}^2$, and the same simulations are repeated to assess the impact of waveguide geometry on the interferometric response. In the third stage, the thermal sensitivity of the imbalanced MZI is investigated for both waveguide geometries. The operating temperature is varied, and its effect on key spectral characteristics including the transmission gain and the free spectral range (FSR) is systematically monitored and compared. In the fourth stage, the effect of the beam splitter coupling ratio is examined by comparing a 100-0% ratio, representing total transmission to one arm, and a 70-30% ratio, representing asymmetric power division between the two arms. In the fifth stage, the designed MZI structures are fabricated on the Applied Nanotools NanoSOI platform using e-beam lithography, and the as-fabricated devices are optically

characterized using an automated swept-laser measurement setup. In the sixth stage, the measured transmission spectra are analyzed through baseline correction, FSR extraction, and corner analysis to extract the group index of the fabricated waveguide and validate it against simulation predictions.

The results demonstrate that the FSR, thermal tuning sensitivity, and fringe visibility of the MZI can be independently controlled through three design parameters: arm length difference, waveguide width, and splitting ratio. Specifically, doubling ΔL from $100 \text{ }\mu\text{m}$ to $200 \text{ }\mu\text{m}$ halves the FSR, widening the waveguide from 500 nm to 600 nm increases the FSR by approximately 3%, and a 3°C temperature increment shifts the transmission spectrum by approximately 1 nm , each independently and without degrading the spectral periodicity of the device. Experimentally, the fabricated MZI3 device ($\Delta L = 200 \text{ }\mu\text{m}$) yielded a measured group index of $n_g = 4.2179 \pm 0.074$ at $\lambda_0 = 1550 \text{ nm}$, deviating from the nominal simulation by only $+0.60\%$ and falling within the corner analysis bounds, confirming that the fabricated waveguide dimensions are consistent with the expected Applied Nanotools process variation.

II. THEORY

A. Mach-Zehnder Interferometer Configurations

A balanced MZI can be modelled as a 2-port device formed by a beam-splitter and a beam-combiner, connected by waveguides with identical geometries and material properties. With this configuration, the signals of the two branches are constructively combined at the MZI output for any wavelength, as illustrated in Figure 1.

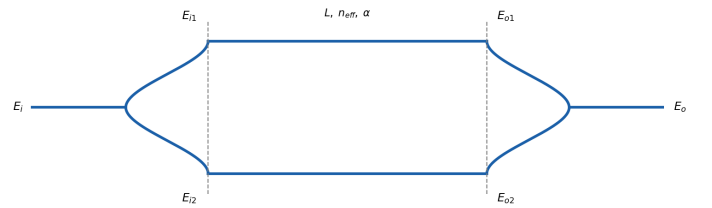


Figure 1. MZI in a balanced configuration. Both branches share the same length L , effective index n_{eff} , and propagation loss α .

To exploit varying levels of interference, a length asymmetry can be introduced between the two branches, as shown in Figure 2. The two waveguides may also exhibit differences in their effective refractive indices ($n_{\text{eff}1}$, $n_{\text{eff}2}$) and attenuation constants (α_1 , α_2). Consequently, the phase mismatch between the two recombined signals produces interference ranging from fully constructive to fully destructive, depending on the signal wavelength.

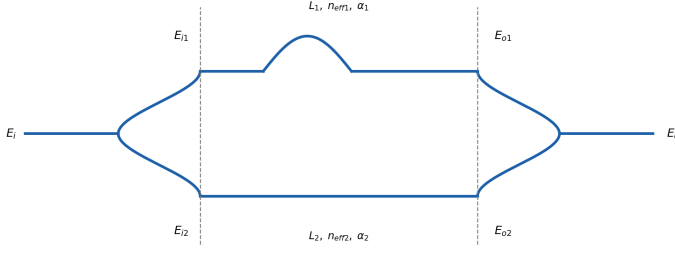


Figure 2. MZI in an imbalanced configuration. The upper and lower branches may have distinct lengths and material properties.

Let E_i denote the input electric field. At the output of the beam-splitter, the complex electric fields entering the upper and lower branches are:

$$E_{i1} = E_{i2} = \frac{1}{\sqrt{2}} E_i \quad (1)$$

After propagating through the respective arms, the fields at the beam-combiner input are:

$$E_{o1} = E_{i1} \cdot e^{-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1} \quad (2)$$

$$E_{o2} = E_{i2} \cdot e^{-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \quad (3)$$

where $\beta_{1,2} = (2\pi/\lambda) n_{eff1,2}$ are the propagation constants. At the beam-combiner:

$$E_o = \frac{1}{\sqrt{2}} (E_{o1} + E_{o2}) \quad (4)$$

Substituting Equations (1)–(3) into Equation (4):

$$E_o = \frac{1}{2} E_i \left(e^{-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \right) \quad (5)$$

The input-output intensity transfer function is:

$$I_o = \frac{1}{4} I_i \left| e^{-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \right|^2 \quad (6)$$

Neglecting propagation losses ($\alpha_1 = \alpha_2 \approx 0$) and assuming equal propagation constants ($\beta_1 = \beta_2 = \beta$), Equation (6) simplifies to:

$$I_o = \frac{1}{2} I_i (1 + \cos(\beta \Delta L)) \quad (7)$$

where $\Delta L = L_1 - L_2$. The free spectral range (FSR) is given by [3]:

$$\text{FSR} = \frac{\lambda^2}{\Delta L \cdot n_g(\lambda)} \quad (8)$$

where n_g is the group index:

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda \cdot \frac{dn_{eff}(\lambda)}{d\lambda} \quad (9)$$

III. SIMULATION

A. Waveguide Mode Analysis: Lumerical MODE

Prior to the MZI transmission simulations, a modal analysis is performed using Lumerical MODE to characterize the spatial field distributions of the fundamental guided modes of the silicon strip waveguides. Two cross-sections are investigated: $500 \times 220 \text{ nm}^2$ and $600 \times 220 \text{ nm}^2$, both at $\lambda = 1550 \text{ nm}$ on a silicon-on-insulator (SOI) platform.

Figure 3 presents the normalized electric field (EF) intensity profiles of the TE and TM modes, and Figure 4 shows the corresponding quasi-TE and quasi-TM profiles obtained from the full vectorial mode solver.

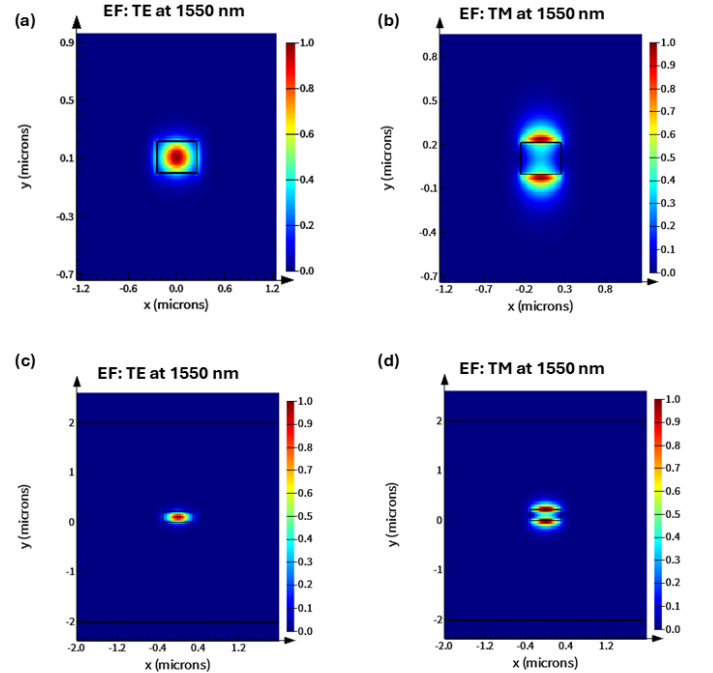


Figure 3. Normalized electric field profiles at 1550 nm: (a) TE mode, $500 \times 220 \text{ nm}^2$; (b) TM mode, $500 \times 220 \text{ nm}^2$; (c) TE mode, $600 \times 220 \text{ nm}^2$; (d) TM mode, $600 \times 220 \text{ nm}^2$.

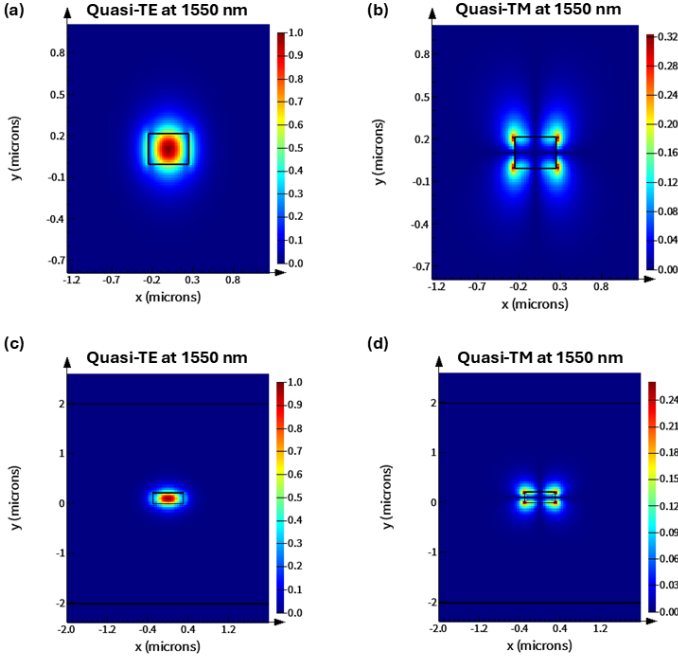


Figure 4. Normalized quasi-TE and quasi-TM profiles at 1550 nm: (a) quasi-TE, $500 \times 220 \text{ nm}^2$; (b) quasi-TM, $500 \times 220 \text{ nm}^2$; (c) quasi-TE, $600 \times 220 \text{ nm}^2$; (d) quasi-TM, $600 \times 220 \text{ nm}^2$.

Two key observations can be made. First, the fundamental quasi-TE mode exhibits strong confinement within the waveguide core in both geometries. Second, increasing the width from 500 nm to 600 nm results in tighter optical confinement of the quasi-TE mode, as evidenced by the more localized intensity distribution in Figures 3(c) and 4(c). In contrast, the TM and quasi-TM modes display a characteristic dual-lobe field distribution extending beyond the waveguide boundaries, indicating weaker confinement. These profiles confirm the selection of the fundamental quasi-TE mode for all subsequent MZI simulations.

B. Effective Index, Group Index and Dispersion

Figure 5 presents the simulated real and imaginary parts of n_{eff} and the group index n_g over 1500–1600 nm for the fundamental quasi-TE mode of both waveguide geometries.

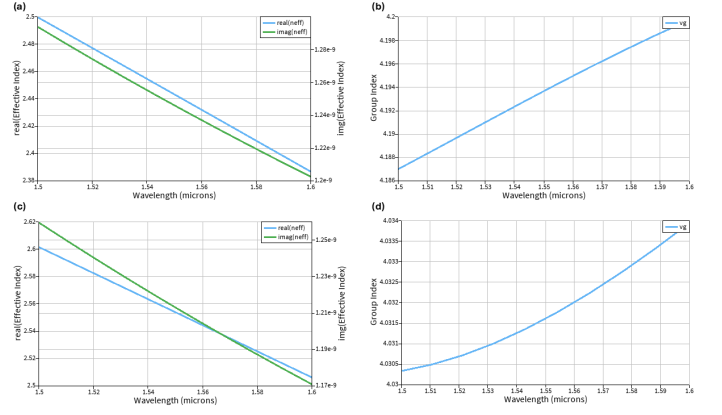


Figure 5. Simulated n_{eff} and n_g versus wavelength: (a) real and imaginary n_{eff} , $500 \times 220 \text{ nm}^2$; (b) n_g , $500 \times 220 \text{ nm}^2$; (c) real and imaginary n_{eff} , $600 \times 220 \text{ nm}^2$; (d) n_g , $600 \times 220 \text{ nm}^2$.

For the $500 \times 220 \text{ nm}^2$ waveguide, the real part of n_{eff} decreases monotonically from 2.50 at 1500 nm to 2.38 at 1600 nm, exhibiting normal dispersion. The group index n_g increases from 4.187 to 4.198 over the same range. For the $600 \times 220 \text{ nm}^2$ waveguide, n_{eff} decreases from 2.60 to 2.50, while n_g ranges from 4.030 to 4.034. In both cases, the imaginary part of n_{eff} remains on the order of 10^{-9} , confirming negligible propagation loss. The higher n_{eff} and lower n_g of the 600 nm waveguide are consistent with its stronger modal confinement, as observed in Section III-A.

C. Compact Waveguide Model

The dispersive behavior of n_{eff} is approximated using a Taylor expansion centred at $\lambda_0 = 1550 \text{ nm}$:

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \quad (10)$$

where n_1 is the effective index at λ_0 , n_2 is the first-order dispersion coefficient, and n_3 is the second-order dispersion coefficient. The coefficients are extracted by fitting Equation (10) to the Lumerical MODE data over 1500–1600 nm and are listed in Table I.

Table I
TAYLOR EXPANSION COEFFICIENTS OF $n_{\text{eff}}(\lambda)$ FOR BOTH STRIP WAVEGUIDES. $\lambda_0 = 1550 \text{ nm}$, QUASI-TE MODE.

Width	n_1	$n_2 (\mu\text{m}^{-1})$	$n_3 (\mu\text{m}^{-2})$
$500 \times 220 \text{ nm}^2$	2.443334	-1.129154	-0.041348
$600 \times 220 \text{ nm}^2$	2.553933	-0.953390	-0.012893

Both fits achieve maximum residuals below 7×10^{-6} , confirming excellent agreement with the simulation data. The coefficient n_1 confirms the higher effective index of the 600 nm waveguide (2.554 vs. 2.443) due to stronger modal confinement. The larger magnitude of n_2 for the 500 nm waveguide reflects its greater wavelength sensitivity, consistent with weaker confinement and larger evanescent field exposure.

D. Thermo-Optic Characterization

The thermo-optic effect of silicon was characterized by simulating the fundamental quasi-TE mode at two temperature conditions using the FDE solver in Lumerical MODE. The temperature-dependent refractive index is modelled as [4]:

$$n(T) = 3.47 + 1.86 \times 10^{-4} \cdot \Delta T \quad (11)$$

The effective modal thermo-optic coefficient is extracted as:

$$\frac{dn_{\text{eff}}}{dT} = \frac{n_{\text{eff}}(10^\circ\text{C}) - n_{\text{eff}}(0^\circ\text{C})}{10} \quad (12)$$

Results are summarized in Table II.

Table II
SIMULATED THERMO-OPTIC COEFFICIENTS FOR BOTH STRIP WAVEGUIDES
AT $\lambda = 1550$ NM.

Width	$n_{\text{eff}}(0^\circ\text{C})$	$n_{\text{eff}}(10^\circ\text{C})$	$\Delta n_{\text{eff}}/\Delta T$ (K^{-1})
$500 \times 220 \text{ nm}^2$	2.43748	2.43946	1.978×10^{-4}
$600 \times 220 \text{ nm}^2$	2.54818	2.55012	1.942×10^{-4}

Both values are in close agreement with the bulk silicon literature value of $1.86 \times 10^{-4} \text{ K}^{-1}$, with deviations of +6.3% and +4.4% respectively. These small positive deviations arise because the simulated value includes contributions from the evanescent field overlap with the SiO_2 cladding. The $500 \times 220 \text{ nm}^2$ waveguide exhibits a slightly higher coefficient due to its lower modal confinement.

The temperature rise ΔT_π required to achieve a π phase shift in the heated MZI arm is:

$$\Delta T_\pi = \frac{\lambda}{2 \cdot \frac{dn_{\text{eff}}}{dT} \cdot L_{\text{heater}}} \quad (13)$$

For $L_{\text{heater}} = 100 \text{ }\mu\text{m}$ at $\lambda = 1550 \text{ nm}$, $\Delta T_\pi \approx 39.2 \text{ K}$ for the 500 nm waveguide and $\Delta T_\pi \approx 39.9 \text{ K}$ for the 600 nm waveguide, both achievable with standard TiN microheaters on the SOI platform.

E. Bend Loss Analysis

The mode overlap between the fundamental quasi-TE mode of the straight and bent waveguide was computed using Lumerical MODE at $\lambda = 1550 \text{ nm}$. The power transmission through a bend is:

$$T_{\text{bend}} = |\text{overlap}|^2, \quad \text{Loss}(\%) = (1 - T_{\text{bend}}) \times 100 \quad (14)$$

A bend loss threshold of 0.1% was adopted as the design criterion. Results are summarized in Table III.

Table III
SIMULATED BEND LOSS FOR BOTH STRIP WAVEGUIDES AT $\lambda = 1550$ NM.
DESIGN THRESHOLD: < 0.1%.

Width	Radius (μm)	Power coupling	T_{bend}	(%)
500 nm	5	0.99949	0.99898	0.260
500 nm	8	0.999471	0.998942	0.106
500 nm	8.5	0.999532	0.999064	0.094
600 nm	5	0.996831	0.993676	0.077

For the $500 \times 220 \text{ nm}^2$ waveguide, the 0.1% threshold is exceeded at $R = 5 \text{ }\mu\text{m}$ (0.260%) and $R = 8 \text{ }\mu\text{m}$ (0.106%). A minimum bend radius of $R = 8.5 \text{ }\mu\text{m}$ is therefore required. For the $600 \times 220 \text{ nm}^2$ waveguide, the threshold is met at $R = 5 \text{ }\mu\text{m}$ (0.077%) due to its stronger modal confinement and more localized field distribution. In summary, minimum bend radii of $8.5 \text{ }\mu\text{m}$ and $5 \text{ }\mu\text{m}$ are adopted for the 500 nm and 600 nm designs respectively.

F. MZI Circuit Simulation: Lumerical INTERCONNECT

The MZI transmission spectra are simulated using Lumerical INTERCONNECT with S-parameter-based component models. The circuit schematic, used for both waveguide geometries, is shown in Figure 6. It consists of two Y-branch splitter/combiner elements (SPAR), two pairs of waveguide segments (WGD) with asymmetric lengths, and an Optical Network Analyzer (ONA). The upper arm has a length of $300 \text{ }\mu\text{m}$ with a loss of 0.09 dB/m , while the lower arm has a length of $100 \text{ }\mu\text{m}$ with a loss of 0.03 dB/m , giving $\Delta L = 200 \text{ }\mu\text{m}$.

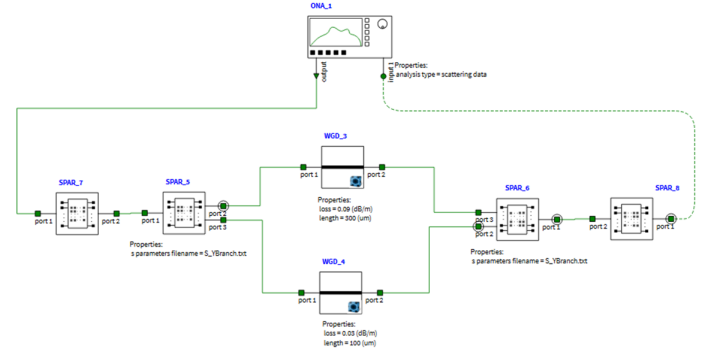


Figure 6. Lumerical INTERCONNECT schematic of the imbalanced MZI ($\Delta L = 200 \text{ }\mu\text{m}$). Upper arm: $300 \text{ }\mu\text{m}$, 0.09 dB/m . Lower arm: $100 \text{ }\mu\text{m}$, 0.03 dB/m .

For thermal tuning studies, a thermo-optic heater element is incorporated into both arms, as shown in Figure 7. The heater induces a temperature-dependent phase shift that translates the transmission spectrum along the wavelength axis.

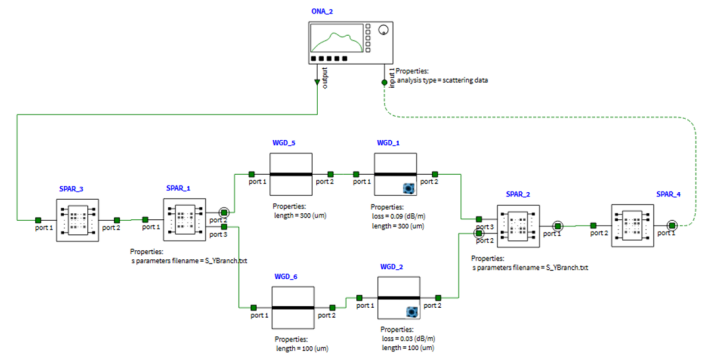


Figure 7. Lumerical INTERCONNECT schematic of the imbalanced MZI with integrated thermo-optic heaters in both arms, used for thermal tuning simulations.

The simulated MZI transmission spectra for the $500 \times 220 \text{ nm}^2$ strip waveguide are evaluated for three configurations: a balanced MZI ($\Delta L = 0 \text{ }\mu\text{m}$) and two imbalanced cases with $\Delta L = 100 \text{ }\mu\text{m}$ and $\Delta L = 200 \text{ }\mu\text{m}$. The balanced MZI exhibits a flat transmission response at 0 dB, confirming fully constructive interference at all wavelengths. For the imbalanced cases, periodic transmission dips appear, with the FSR decreasing as ΔL increases. The $\Delta L = 100 \text{ }\mu\text{m}$ case produces a wider FSR with shallower extinction, while $\Delta L = 200 \text{ }\mu\text{m}$ yields a narrower FSR with deeper nulls exceeding -40 dB . The extracted TE FSR ranges from $5.4\text{--}6.1 \text{ nm}$ ($\Delta L = 100 \text{ }\mu\text{m}$) and $2.7\text{--}3.1 \text{ nm}$ ($\Delta L = 200 \text{ }\mu\text{m}$) over $1500\text{--}1600 \text{ nm}$. Thermal tuning at $\Delta L = 200 \text{ }\mu\text{m}$ produces a red-shift of approximately 1 nm per 3°C , while the FSR remains stable. The same trends are observed for the $600 \times 220 \text{ nm}^2$ waveguide, with slightly larger FSR values of $5.6\text{--}6.3 \text{ nm}$ and $2.8\text{--}3.2 \text{ nm}$ due to its lower group index. A full quantitative comparison is presented in Section IV.

1) *Design Parameter Summary:* Table IV summarises all MZI configurations investigated, with the theoretically expected FSR at $\lambda = 1550 \text{ nm}$ and the temperature conditions applied.

Table IV
MZI DESIGN PARAMETER VARIATIONS, EXPECTED FSR AT $\lambda = 1550 \text{ nm}$,
AND THERMAL TUNING CONDITIONS.

Design	Width	Height	$\Delta L \text{ (}\mu\text{m)}$	FSR (nm)	Temp. ($^\circ\text{C}$)
A1	500 nm	220 nm	0	—	—
A2	500 nm	220 nm	100	5.7	—
A3	500 nm	220 nm	200	2.9	26, 29, 32
A4	600 nm	220 nm	0	—	—
A5	600 nm	220 nm	100	5.9	—
A6	600 nm	220 nm	200	3.0	26, 29, 32

Designs A1 and A4 are balanced reference configurations. Designs A2, A3, A5, and A6 are imbalanced. Thermal tuning is applied to A3 and A6 ($\Delta L = 200 \text{ }\mu\text{m}$) at three temperatures: 26°C , 29°C , and 32°C , achievable with standard TiN microheaters.

IV. SIMULATION RESULTS AND DISCUSSION

Building on the individual waveguide and circuit simulations described in Section III, this section presents a direct quantitative comparison across all MZI configurations, highlighting the influence of ΔL , waveguide width, temperature, and splitting ratio on the spectral response.

A. Transmission Gain Comparison

Figure 8 presents the MZI transmission spectra over $1540\text{--}1560 \text{ nm}$ for all four imbalanced configurations: the $500 \times 220 \text{ nm}^2$ and $600 \times 220 \text{ nm}^2$ waveguides, each at $\Delta L = 100 \text{ }\mu\text{m}$ and $\Delta L = 200 \text{ }\mu\text{m}$.

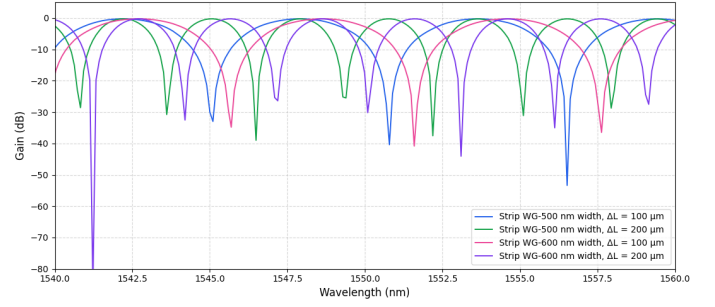


Figure 8. Simulated MZI transmission spectra for the $500 \times 220 \text{ nm}^2$ and $600 \times 220 \text{ nm}^2$ strip waveguides at $\Delta L = 100 \text{ }\mu\text{m}$ and $\Delta L = 200 \text{ }\mu\text{m}$.

The balanced MZI ($\Delta L = 0 \text{ }\mu\text{m}$) exhibits a flat transmission response at 0 dB across the entire wavelength range, which indicates that the two arms contribute constructively at all wavelengths. For the imbalanced cases, periodic transmission dips appear due to destructive interference, with the spacing between consecutive minima the free spectral range (FSR) decreasing as ΔL increases. Specifically, the $\Delta L = 100 \text{ }\mu\text{m}$ case produces a wider FSR with shallower extinction, while the $\Delta L = 200 \text{ }\mu\text{m}$ case yields a narrower FSR with deeper and more pronounced transmission nulls, reaching extinction depths beyond -40 dB at certain wavelengths.

The $600 \times 220 \text{ nm}^2$ geometry follows the same qualitative trends as the $500 \times 220 \text{ nm}^2$ case. However, for the same ΔL , the 600 nm waveguide produces a slightly wider FSR, attributable to its lower group index n_g (4.032 vs. 4.194), as predicted by Equation (8). The extinction depth at the transmission nulls varies between configurations due to the wavelength dependence of the propagation constant mismatch.

B. Free Spectral Range (FSR) Comparison

Table V summarises the simulated FSR values at $\lambda = 1550 \text{ nm}$, and Figure 9 shows the extracted TE FSR as a function of wavelength over $1500\text{--}1600 \text{ nm}$ for all four configurations.

Table V
SIMULATED TE FSR AT $\lambda = 1550 \text{ nm}$.

Width	$\Delta L \text{ (}\mu\text{m)}$	FSR @ 1550 nm
$500 \times 220 \text{ nm}^2$	100	5.7
	200	2.9
$600 \times 220 \text{ nm}^2$	100	5.9
	200	3.0

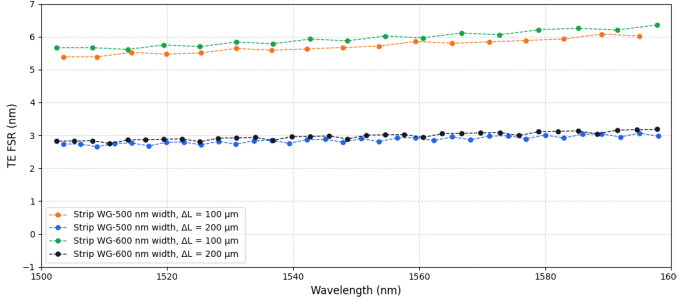


Figure 9. Extracted TE FSR versus wavelength for both strip waveguides at $\Delta L = 100 \mu\text{m}$ and $\Delta L = 200 \mu\text{m}$ over 1500–1600 nm.

The balanced MZI yields an FSR of 0 nm, consistent with the absence of periodic spectral features. For $\Delta L = 100 \mu\text{m}$, the FSR of the 500 nm waveguide ranges from approximately 5.4 nm to 6.1 nm across the observed wavelength window, exhibiting a slight monotonic increase with wavelength as predicted by Equation (8). For $\Delta L = 200 \mu\text{m}$, the FSR is approximately halved, ranging from 2.7 nm to 3.1 nm, consistent with the inverse proportionality between FSR and ΔL .

For the $600 \times 220 \text{ nm}^2$ waveguide, the wider core results in a slightly larger FSR for the same ΔL , due to its lower group index n_g (4.032 vs. 4.194). For $\Delta L = 100 \mu\text{m}$, the FSR ranges from approximately 5.6 nm to 6.3 nm, while for $\Delta L = 200 \mu\text{m}$ it ranges from 2.8 nm to 3.2 nm across the 1500–1600 nm window. In all cases, the FSR increases monotonically with wavelength, consistent with the λ^2 dependence in Equation (8). The consistently larger FSR of the 600 nm waveguide confirms that waveguide width is an effective parameter for tuning the spectral periodicity of the MZI.

C. Thermal Tuning Comparison

Figure 10 presents the thermal tuning response of the imbalanced MZI at $\Delta L = 200 \mu\text{m}$ for both waveguide geometries at three operating temperatures: 26°C, 29°C, and 32°C.

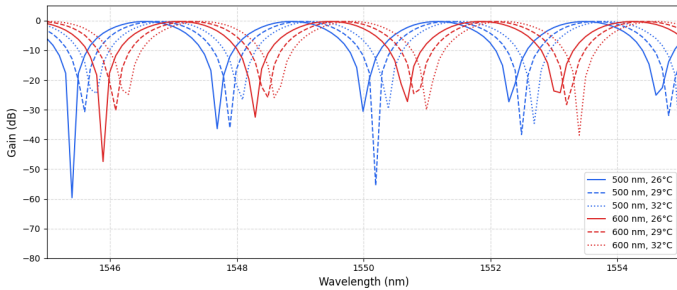


Figure 10. Thermal tuning of the MZI transmission spectrum for the $500 \times 220 \text{ nm}^2$ (blue) and $600 \times 220 \text{ nm}^2$ (red) strip waveguides at $\Delta L = 200 \mu\text{m}$, at 26°C, 29°C, and 32°C.

Increasing the operating temperature produces a progressive red-shift of the transmission spectrum in both waveguide

geometries, consistent with the positive thermo-optic coefficient of silicon ($dn/dT \approx 1.86 \times 10^{-4} \text{ K}^{-1}$). Each 3°C increment shifts the spectral dips toward longer wavelengths, while the FSR and overall spectral shape remain largely unchanged. This confirms that thermal tuning via the thermo-optic effect provides an effective mechanism for wavelength-selective control of the MZI transmission without degrading its spectral periodicity.

A comparison between the two geometries reveals that the 500 nm waveguide exhibits a marginally larger wavelength shift per degree Celsius than the 600 nm waveguide, consistent with its higher simulated modal thermo-optic coefficient ($1.978 \times 10^{-4} \text{ K}^{-1}$ vs. $1.942 \times 10^{-4} \text{ K}^{-1}$, Table II). For the $600 \times 220 \text{ nm}^2$ waveguide, increasing temperature similarly induces a red-shift, though the magnitude per degree Celsius is marginally smaller, consistent with its lower modal thermo-optic coefficient. In both cases, the FSR remains stable across all temperatures, which verifies that the thermal tuning shifts the operating point of the MZI without altering its fundamental spectral periodicity.

D. Effect of Beam Splitter Coupling Ratio

The effect of the beam splitter coupling ratio is investigated by comparing a 100-0% splitting ratio, in which all power is directed into a single arm, and a 70-30% ratio, representing asymmetric power division. Both conditions are implemented in Lumerical INTERCONNECT by modifying the Y-branch S-parameter file, with all other parameters identical to the imbalanced MZI at $\Delta L = 200 \mu\text{m}$.

Figure 11 presents the simulated MZI transmission for splitting ratios of 100-0% and 70-30% at $\Delta L = 200 \mu\text{m}$.

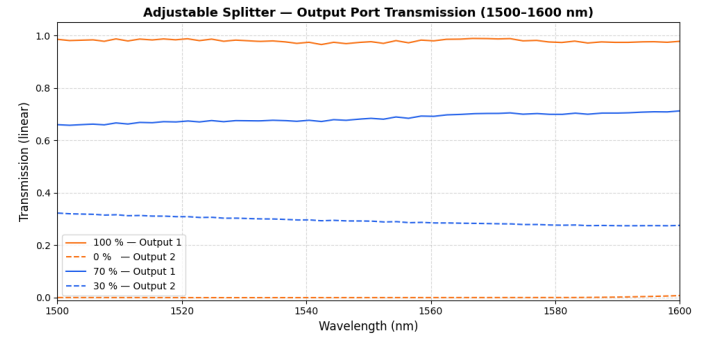


Figure 11. Simulated MZI transmission for splitting ratios of 100-0% and 70-30% at $\Delta L = 200 \mu\text{m}$.

For the 100-0% splitting ratio, all input power is directed into a single arm, resulting in no interference at the output and a flat transmission response. For the 70-30% ratio, the power imbalance between the two arms reduces the visibility of the interference fringes compared to the ideal 50-50% case. The extinction ratio of the transmission dips decreases, as incomplete destructive interference prevents full cancellation at the output. The FSR, however, remains unchanged, since it is determined solely by ΔL and n_g , both of which are unaffected by the splitting ratio.

E. Group Index Extraction from FSR

Rearranging Equation (8), the group index can be extracted directly from the measured or simulated FSR as:

$$n_g(\lambda) = \frac{\lambda^2}{\text{FSR}(\lambda) \cdot \Delta L} \quad (15)$$

where λ , FSR, and ΔL are all expressed in micrometers. Table VI compares the extracted n_g values against the directly simulated values at $\lambda = 1550$ nm for both waveguide geometries.

Table VI
GROUP INDEX EXTRACTED FROM SIMULATED FSR AT $\lambda = 1550$ nm,
COMPARED TO DIRECTLY SIMULATED n_g .

Width	ΔL (μm)	n_g extracted	n_g simulated
$500 \times 220 \text{ nm}^2$	100	4.2149	4.1942
	200	4.1422	
$600 \times 220 \text{ nm}^2$	100	4.0720	4.0318
	200	4.0042	

The simulated n_g values of 4.1942 and 4.0318 represent the group index computed directly from the Lumerical MODE frequency sweep at $\lambda = 1550$ nm, and are common to both ΔL cases for each waveguide width, since n_g is a property of the waveguide geometry alone, and it is independent of the MZI arm length difference.

All extracted values agree with the directly simulated n_g to within 1.3%, verifying that Equation (15) provides a reliable and experimentally accessible route to characterizing n_g directly from the transmission spectrum without requiring a direct mode simulation. This procedure was applied to the measured MZI3 spectrum, yielding $n_g = 4.2179$.

V. LAYOUT

The physical layout of the MZI designs was implemented using KLayout with the SiEPIC EBeam Process Design Kit (PDK), exclusively for the $500 \times 220 \text{ nm}^2$ strip waveguide geometry. The complete layout is shown in Figure 12.

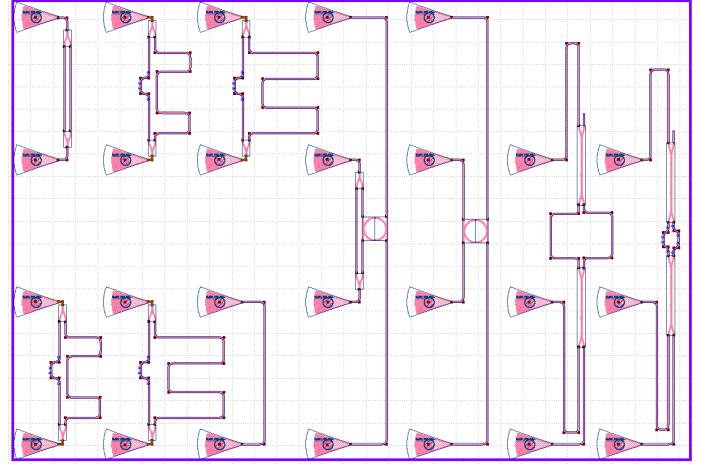


Figure 12. KLayout view of the complete fabrication layout containing ten photonic structures arranged across two rows. Top row (left to right): balanced MZI, imbalanced MZI ($\Delta L = 100 \mu\text{m}$), imbalanced MZI ($\Delta L = 200 \mu\text{m}$), ring resonator coupled to MZI, double-pass ring resonator, adjustable splitter (100:0), and adjustable splitter ($\approx 70:30$). Bottom row (left to right): imbalanced MZI ($\Delta L = 100 \mu\text{m}$), imbalanced MZI ($\Delta L = 200 \mu\text{m}$), and U-loop reference structure. Grating coupler pitch: $127 \mu\text{m}$.

A. Layout Overview

The layout contains ten photonic structures arranged in two rows. All structures use $500 \times 220 \text{ nm}^2$ strip waveguides with a minimum bend radius of $8.5 \mu\text{m}$, as determined from the bend loss analysis in Section III. The structures are described below.

Top Row:

- **MZI 1: Balanced MZI ($\Delta L = 0 \mu\text{m}$):** The reference structure with equal arm lengths. Both arms share identical waveguide geometry and propagation constants, producing fully constructive interference at all wavelengths and a flat transmission response at 0 dB. This structure serves as the baseline for validating the Y-branch insertion loss and grating coupler response independently of interferometric effects.
- **MZI 2: Imbalanced MZI ($\Delta L = 100 \mu\text{m}$):** An asymmetric MZI in which the upper arm is extended by $100 \mu\text{m}$ relative to the lower arm. The expected FSR at $\lambda = 1550$ nm is approximately 5.7 nm , as predicted by Equation (8) using the simulated group index $n_g = 4.194$.
- **MZI 3: Imbalanced MZI ($\Delta L = 200 \mu\text{m}$):** An asymmetric MZI with a $200 \mu\text{m}$ arm length difference, producing an FSR of approximately 2.9 nm — approximately half that of MZI 2 — consistent with the inverse proportionality between FSR and ΔL in Equation (8).
- **Ring Resonator coupled to MZI:** A ring resonator integrated into one arm of an imbalanced MZI. The ring introduces additional wavelength-selective phase shifts into the MZI transfer function, enabling investigation of the combined spectral response of two coupled interferometric elements.
- **Double-Pass Ring Resonator:** A ring resonator configured for double-pass operation, in which the optical field circulates through the ring twice per round trip.

This structure allows characterization of the enhanced wavelength selectivity and sharpened resonance linewidth achievable with double-pass coupling geometries.

- **Adjustable Splitter (100:0 splitting):** A Y-branch-based adjustable splitter configured with the coupling geometry biased towards the longer arm, resulting in approximately 100:0 power splitting. In this configuration, all input power is directed into a single MZI arm, eliminating interference at the output and producing a flat transmission response. This structure provides a reference for the single-arm propagation loss and confirms the absence of interferometric fringes under complete power imbalance.
- **Adjustable Splitter ($\approx 70:30$ splitting):** A Y-branch-based adjustable splitter configured with asymmetric coupling, directing approximately 70% of the input power into one arm and 30% into the other. This power imbalance reduces the fringe visibility of the MZI output relative to the ideal 50:50 case, as incomplete destructive interference at the combiner prevents full cancellation. The FSR is unaffected, since it is determined solely by ΔL and n_g .

Bottom Row:

- **MZI 4 — Imbalanced MZI ($\Delta L = 100 \mu\text{m}$):** A duplicate of MZI 2, included to assess fabrication repeatability and quantify within-chip manufacturing variability across nominally identical designs.
- **MZI 5 — Imbalanced MZI ($\Delta L = 200 \mu\text{m}$):** A duplicate of MZI 3, similarly included for repeatability assessment and statistical comparison of the FSR and peak wavelength against MZI 3.
- **U-Loop Reference Structure:** Two grating couplers connected by a continuous strip waveguide forming a U-shaped loop. This structure contains no splitter, combiner, or interferometric element, and serves as a direct reference for characterizing the wavelength-dependent insertion loss and spectral envelope of the grating couplers. The measured transmission of this structure is used to normalize the MZI spectra during post-processing of experimental data.

B. Port Labelling and Input Convention

All input ports in the layout follow the SiEPIC EBeam naming convention:

`opt_in_TE_1550_device_harithak_YOURDEVICE`

where `harithak` identifies the designer and `YOURDEVICE` is replaced by the specific structure identifier for each design (e.g., `MZI1`, `MZI2`, `MZI3`). This labelling format is required by the SiEPIC automated measurement system to correctly identify and route the optical probe to each input port during automated testing.

C. Circuit Simulation Verification

Prior to submitting the layout for fabrication, circuit-level simulations were performed on each MZI structure directly within the layout environment to verify the expected interferometric behavior. A continuous-wave laser source was

placed at the input port, and individual photodetectors were connected to each output port. The simulated transmission spectra were extracted and compared against the reference INTERCONNECT simulations presented in Section III.

The circuit simulation results for all three MZI configurations : balanced ($\Delta L = 0 \mu\text{m}$), imbalanced ($\Delta L = 100 \mu\text{m}$), and imbalanced ($\Delta L = 200 \mu\text{m}$), were found to be in close agreement with the expected behavior, confirming the correctness of the waveguide routing, arm length differences, and Y-branch connections implemented in the layout.

D. Design Rule Check and Coordinate Extraction

Following the circuit simulations, two automated verification steps were performed:

- 1) **Automated Coordinate Extraction:** The port labels of all eight structures were automatically extracted and verified against the expected naming convention. All labels were correctly detected, confirming that the measurement system will be able to identify and address each input port during automated probing.
- 2) **Design Rule Check (DRC):** A full DRC was executed to identify any layout violations, including minimum spacing, minimum width, and waveguide bend radius constraints. No errors were detected, confirming that the layout is compliant with the SiEPIC EBeam fabrication design rules.

VI. FABRICATION

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (Edmonton, Canada), which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device layer thickness, and $2 \mu\text{m}$ buffer oxide thickness were used as the base material. The wafer was pre-diced into $25 \text{ mm} \times 25 \text{ mm}$ square substrates, with scribe lines on the backside to facilitate easy separation into smaller chips once fabrication was complete.

After an initial wafer clean using piranha solution (3:1 $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$) for 15 minutes, followed by deionized water and isopropanol (IPA) rinses, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dose was corrected for proximity effects arising from backscattered electrons from nearby features, and the shape-writing order was optimized for efficient patterning and minimal beam drift. After EBL exposure, the resist was developed in a tetramethylammonium hydroxide (TMAH) solution, and the chips were subsequently inspected optically for residues and defects.

The chips were then mounted on a 4-inch handle wafer and subjected to an anisotropic inductively coupled plasma reactive-ion etch (ICP-RIE) using chlorine chemistry, following qualification of the etch rate. Residual HSQ resist was removed using a 10:1 buffered oxide etch (BOE), and the patterned devices were inspected by scanning electron

microscopy (SEM) to verify patterning fidelity and etch quality. Finally, a 2.2 μm silicon dioxide cladding was deposited by plasma-enhanced chemical vapour deposition (PECVD) using a tetraethyl orthosilicate (TEOS) precursor at 300 $^{\circ}\text{C}$. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide, and cladding thicknesses prior to delivery.

VII. EXPERIMENTAL METHODS

To characterize the devices, a custom-built automated test setup [3] with automated control software written in Python was used [5]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization-maintaining (PM) fiber was used to maintain the polarization state of the light and to couple the TE polarization into the grating couplers [6]. A 90 $^{\circ}$ rotation of the PM fiber was applied to inject light into the TM grating couplers [6]. A polarization-maintaining fiber array was used to couple light in and out of the chip [7].

VIII. ANALYSIS

A. Corner Analysis of a 500 nm Strip Waveguide MZI

Manufacturing variability in silicon photonics arises primarily from two sources: silicon layer thickness variation and waveguide width variation introduced during lithography and etching. To evaluate their combined impact on circuit performance, a corner analysis was performed on an unbalanced Mach-Zehnder Interferometer (MZI) with a path length difference of $\Delta L = 200 \mu\text{m}$, based on a 500 nm $\times$ 220 nm silicon strip waveguide operating in the TE polarization at $\lambda_0 = 1550 \text{ nm}$. This matches the geometry of the fabricated MZI3 device characterized in Section VIII-B.

The silicon layer thickness was taken from the Soitec wafer specification as $219.2 \pm 3.9 \text{ nm}$ (1σ), giving an expected range of 215.3 nm to 223.1 nm. The waveguide width variation was estimated from observed e-beam lithography process behavior, ranging from -30 nm to $+10 \text{ nm}$ relative to the nominal 500 nm target, giving a width range of 470 nm to 510 nm. Five waveguide geometries were simulated using the Finite Difference Eigenmode (FDE) solver in Lumerical MODE: one nominal design and four corners representing the extreme combinations of these two parameters, as summarized in Table VII.

Table VII

CORNER ANALYSIS: WAVEGUIDE GEOMETRIES AND EXTRACTED OPTICAL PARAMETERS AT $\lambda = 1550 \text{ nm}$, TE POLARIZATION. FSR COMPUTED FOR $\Delta L = 200 \mu\text{m}$.

Corner	Width (nm)	Thickness (nm)	n_{eff}	n_g	FSR (nm)
Nominal	500	219.2	2.4403	4.1927	2.865
FF	510	223.1	2.4691	4.1825	2.872
SS	470	215.3	2.3711	4.2463	2.829
SF	470	223.1	2.4006	4.2571	2.822
FS	510	215.3	2.4397	4.1734	2.878

For each corner, a wavelength sweep from 1550 nm to 1600 nm was performed to extract the waveguide dispersion. The effective index $n_{\text{eff}}(\lambda)$ was fitted to a second-order

polynomial compact model of the form given in Equation (10). The group index was then obtained analytically from the polynomial coefficients using Equation (9). These compact models were subsequently used to simulate the MZI transmission spectrum for each corner via the lossless transfer function:

$$T(\lambda) = \cos^2\left(\frac{\pi n_{\text{eff}}(\lambda) \Delta L}{\lambda}\right) \quad (16)$$

The resulting transmission spectra are shown in Figure 13. The full spectra (1520–1575 nm) are shown in the upper panel, with a zoomed view in the lower panel to clearly resolve the corner-dependent peak wavelength shifts.

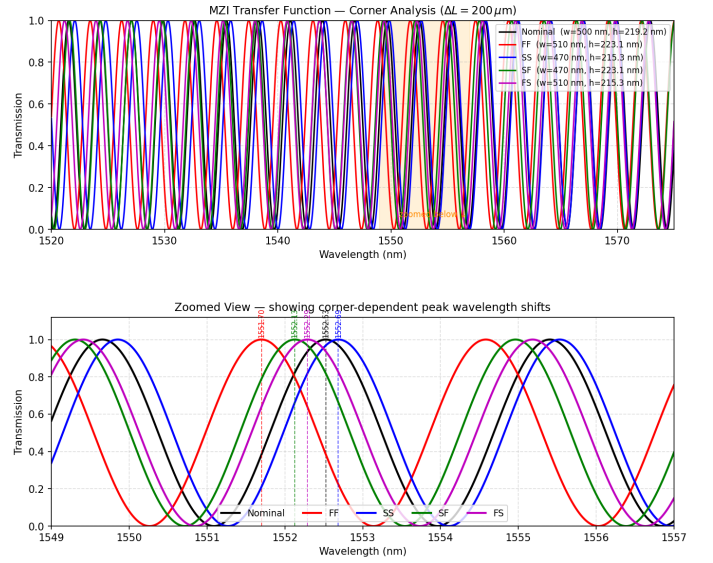


Figure 13. MZI transfer function for the five corner analysis cases ($\Delta L = 200 \mu\text{m}$, 500 $\times$ 220 nm TE strip waveguide). Upper panel: full spectrum from 1520 to 1575 nm. Lower panel: zoomed view showing peak wavelength shifts across corners.

The group index at 1550 nm ranged from a minimum of $n_g = 4.1734$ at the FS corner (wide and thin: $w = 510 \text{ nm}$, $h = 215.3 \text{ nm}$) to a maximum of $n_g = 4.2571$ at the SF corner (narrow and thick: $w = 470 \text{ nm}$, $h = 223.1 \text{ nm}$). This behavior is physically consistent: a narrower waveguide confines the TE mode less tightly, increasing the evanescent field fraction and hence the waveguide dispersion, which raises the group index. The effective index showed a wider variation, ranging from $n_{\text{eff}} = 2.371$ (SS corner) to $n_{\text{eff}} = 2.469$ (FF corner).

Using Equation (8), the FSR ranges from a minimum of 2.822 nm at the SF corner (highest n_g) to a maximum of 2.878 nm at the FS corner (lowest n_g). The total FSR variation of 0.057 nm represents approximately 2.0% of the nominal FSR, which is relatively small. In contrast, the peak transmission wavelength shifts by up to 3.2 nm from the nominal position, with the largest deviations occurring at the narrow-waveguide corners (SS and SF), where the reduced n_{eff} alters the phase accumulated in the longer MZI arm. Such a shift is significant in wavelength-sensitive applications and would require active thermal tuning to compensate, demonstrating the

practical importance of incorporating manufacturing variability into the photonic design flow.

Note that the n_g bounds [4.1734, 4.2571] are independent of ΔL , since n_g is a property of the waveguide cross-section alone. These bounds are therefore directly applicable to the experimental comparison presented in Section VIII-B.

B. Baseline Correction and Spectral Analysis

The raw transmission spectrum of MZI3 ($\Delta L = 200 \mu\text{m}$) exhibits a bell-shaped spectral envelope imposed by the finite bandwidth of the grating couplers [6]. Prior to analysis, this envelope was removed by fitting a fourth-order polynomial to the raw spectrum and subtracting it, shifting the maximum to 0 dB, as shown in Figure 14. This baseline correction flattens the interferometric fringe pattern and enables reliable extraction of the free spectral range, following the method described in [3].

Transmission minima were identified on the corrected spectrum using an automated peak-finding algorithm with a minimum prominence of 1.5 dB and a minimum separation of 150 data points ($\approx 1.2 \text{ nm}$), as shown in Figure 15. A total of 19 minima were detected over the wavelength range 1521–1573 nm, yielding 18 consecutive FSR values ranging from 2.768 nm to 2.936 nm. The median FSR of $2.848 \pm 0.050 \text{ nm}$ was used as the robust estimator, giving a measured group index of:

$$n_g = \frac{\lambda^2}{\text{FSR} \cdot \Delta L} = \frac{(1.550)^2}{2.848 \times 10^{-3} \times 200} = 4.2179 \quad (17)$$

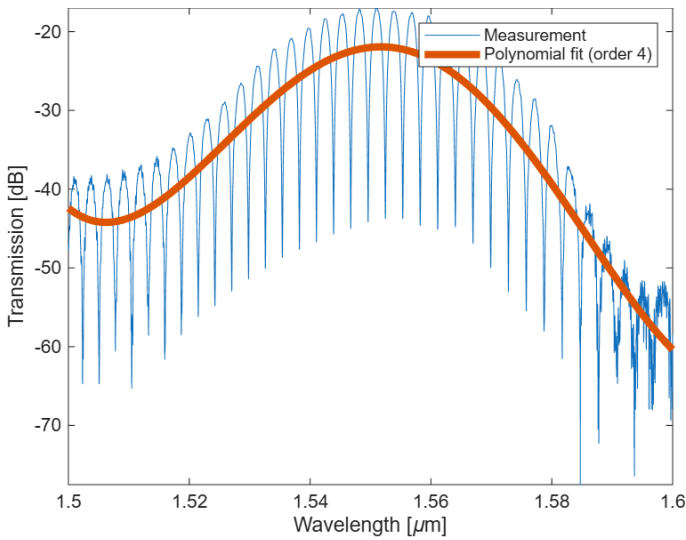


Figure 14. Measured transmission spectrum of MZI3 ($\Delta L = 200 \mu\text{m}$) with the 4th-order polynomial baseline fit used to remove the grating coupler spectral envelope.

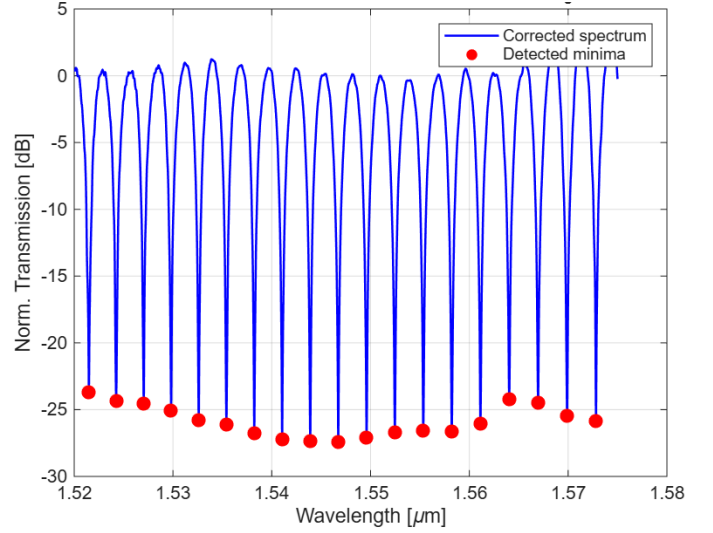


Figure 15. Baseline-corrected transmission spectrum of MZI3 with detected transmission minima (red dots). The median FSR of 2.848 nm yields a measured group index of $n_g = 4.2179$.

C. Measurement versus Simulation

Figure 16 overlays the raw measured transmission spectrum of MZI3 with the simulated transmission spectrum derived from the compact waveguide model (Table I). The bell-shaped envelope visible in the measurement reflects the finite bandwidth of the grating couplers [6], which is removed by the baseline correction described above and shown in Figure 14. Despite the envelope, the fringe periodicity of the two spectra is in close visual agreement across the full 1500–1600 nm wavelength range, confirming that the arm length difference $\Delta L = 200 \mu\text{m}$ was correctly implemented in both the layout and the simulation.

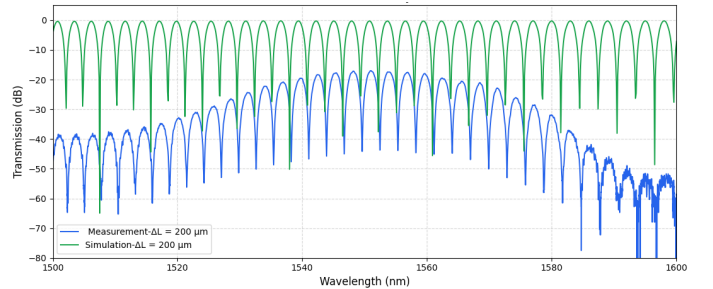


Figure 16. Transmission spectra of MZI3 ($\Delta L = 200 \mu\text{m}$): raw measurement (blue) overlaid with simulation (green). The bell-shaped envelope of the measurement reflects the finite bandwidth of the grating couplers, which is absent from the lossless simulation model. The fringe periodicity is in close agreement across the full 1500–1600 nm range.

The simulated FSR at $\lambda = 1550 \text{ nm}$ is 2.900 nm, compared to the measured value of 2.848 nm - a difference of 1.8%. The measured group index $n_g = 4.2179 \pm 0.074$ deviates from the nominal simulated value of 4.1927 by +0.60%, and falls within the corner analysis bounds of [4.1734, 4.2571], confirming that the fabricated waveguide dimensions are consistent with the expected Applied Nanotools process variation.

Two further discrepancies are noted. First, the measured extinction ratio (≈ 20 – 25 dB) is lower than the simulated value (> 40 dB), attributable to propagation loss and residual Y-branch splitting imbalance not included in the lossless simulation model. Second, the measured transmission peaks are shifted by approximately 0.5 – 1.0 nm toward longer wavelengths relative to the simulation, consistent with the slightly higher measured n_g .

D. Group Index Comparison

The measured group index, extracted from the free spectral range of the baseline-corrected MZI3 transmission spectrum ($\Delta L = 200 \mu\text{m}$), is compared against the simulation predictions and the corner analysis bounds. The group index was computed per fringe pair using $n_g = \lambda^2 / (\text{FSR} \cdot \Delta L)$, yielding a median value of $n_g = 4.2179$ at $\lambda_0 = 1550$ nm. The simulated group index for the nominal $500 \text{ nm} \times 219.2 \text{ nm}$ waveguide geometry, derived from the compact Taylor expansion model, is $n_g = 4.1927$. The measured value deviates from the nominal simulation by $+0.0252$ ($+0.60\%$), which is within the expected fabrication tolerance.

The corner analysis, performed over the width range 470 – 510 nm and thickness range 215.3 – 223.1 nm, yields bounds of $n_g \in [4.1734, 4.2571]$. The measured $n_g = 4.2179$ falls within these bounds, confirming that the fabricated waveguide dimensions are consistent with the Applied Nanotools NanoSOI process variation. The slight positive deviation from the nominal value is consistent with a marginally narrower or thicker waveguide, corresponding to the SF corner ($w = 470$ nm, $h = 223.1$ nm), which is physically plausible given the -30 nm to $+10$ nm width variation and ± 3.9 nm thickness variation expected from e-beam lithography.

IX. CONCLUSION

This work presented a systematic numerical and experimental investigation of the silicon-based Mach-Zehnder Interferometer across multiple design configurations. The fundamental quasi-TE mode was characterised for $500 \times 220 \text{ nm}^2$ and $600 \times 220 \text{ nm}^2$ strip waveguides using Lumerical MODE, confirming that increasing the waveguide width enhances modal confinement and reduces the group index. Compact Taylor-series models were derived for both geometries, achieving fit residuals below 7×10^{-6} .

INTERCONNECT circuit simulations demonstrated that the FSR scales inversely with ΔL and is consistently larger for the 600 nm waveguide due to its lower n_g . Thermal tuning via the thermo-optic effect produced a stable red-shift of approximately 1 nm per 3°C in both geometries, with the FSR remaining unchanged. The 500 nm waveguide exhibited a marginally larger thermal shift, consistent with its higher modal thermo-optic coefficient. The splitting ratio study confirmed that a 70 – 30% ratio reduces fringe visibility without affecting the FSR, while a 100 – 0% ratio eliminates interference entirely.

A corner analysis evaluated the combined effect of silicon layer thickness variation (219.2 ± 3.9 nm, 1σ) and e-beam lithography width variation (-30 nm to $+10$ nm) on the MZI3

spectral response ($\Delta L = 200 \mu\text{m}$). The FSR variation across all corners was found to be small (approximately 2.0% of the nominal value, spanning 2.822 – 2.878 nm), while the peak transmission wavelength shifted by up to 3.2 nm, highlighting the need for active thermal compensation in wavelength-sensitive applications.

The group index extraction method was applied to the measured MZI3 transmission spectrum ($\Delta L = 200 \mu\text{m}$), yielding $n_g = 4.2179$ at $\lambda_0 = 1550$ nm. This value deviates from the nominal simulation by only $+0.60\%$ and falls within the corner analysis bounds $[4.1734, 4.2571]$, confirming that the fabricated waveguide dimensions are consistent with the Applied Nanotools NanoSOI process variation. The measured FSR of 2.848 nm agrees with the simulated value of 2.900 nm to within 1.8% , demonstrating close agreement between the design, simulation, and fabricated device.

X. ACKNOWLEDGEMENT

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